ITA06020 MACHINE LEARNING

Experiment 20 – Future Sales Prediction Using Linear Regression

Aim:
To predict future sales based on historical sales data using the Linear Regression algorithm.

Procedure:

1. Import the required Python libraries.
2. Load the Online Retail dataset.
3. Clean the dataset by removing missing and invalid values.
4. Calculate monthly sales.
5. Train a Linear Regression model using historical sales.
6. Predict sales for the next three months.

Dataset: Online Retail Dataset from the UCI Machine Learning Repository. It contains real-world retail transaction details such as invoice date, quantity and unit price.

Python Code:

```
import pandas as pd

import zipfile

import urllib.request

from sklearn.linear_model import LinearRegression

import matplotlib.pyplot as plt

url = "https://archive.ics.uci.edu/static/public/352/online+retail.zip"

urllib.request.urlretrieve(url, "retail.zip")

with zipfile.ZipFile("retail.zip", "r") as zip_file:

    zip_file.extractall("retail_data")

data = pd.read_excel("retail_data/Online Retail.xlsx")

data = data.dropna(subset=["InvoiceDate"])

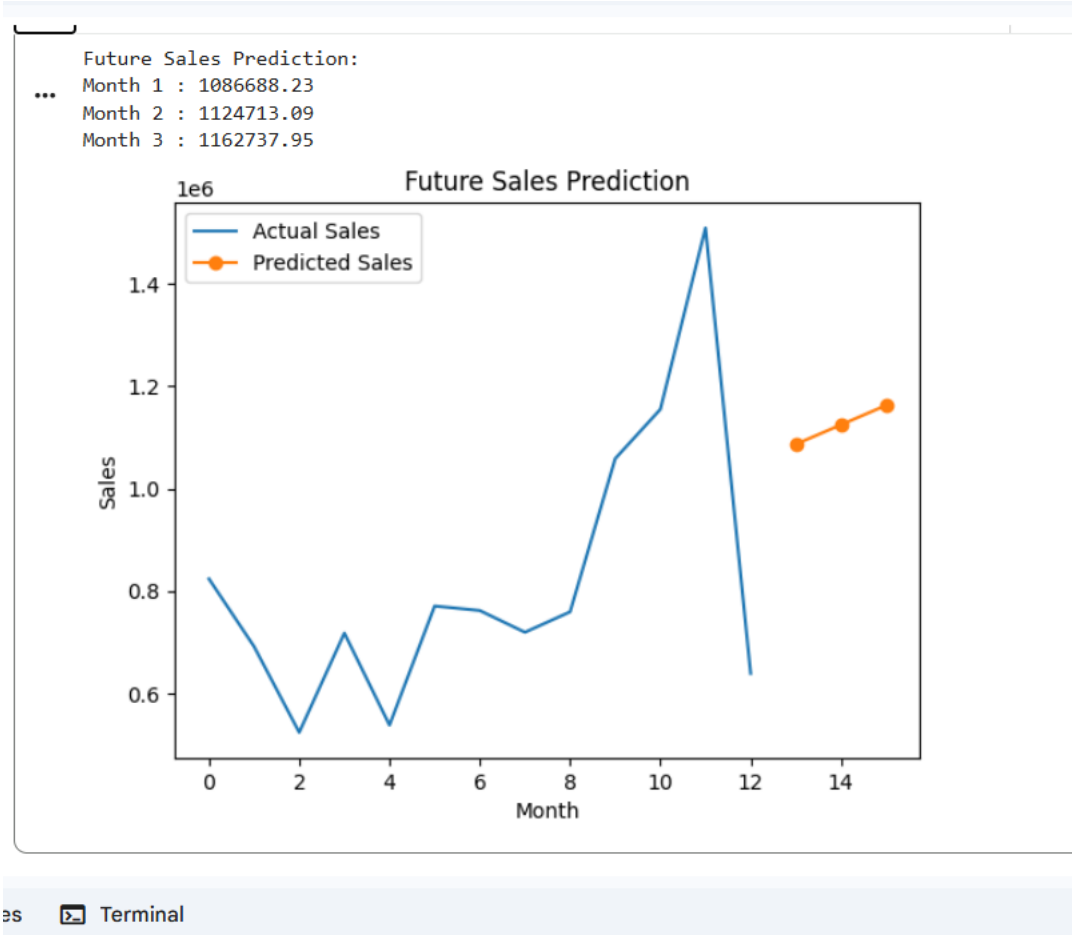
data = data[(data["Quantity"] > 0) & (data["UnitPrice"] > 0)]

data["Sales"] = data["Quantity"] * data["UnitPrice"]
```

```
data["InvoiceDate"] = pd.to_datetime(data["InvoiceDate"])
monthly_sales = data.set_index("InvoiceDate").resample("MS")["Sales"].sum()
X = list(range(len(monthly_sales)))
X = pd.DataFrame(X, columns=["Month"])
y = monthly_sales.values
model = LinearRegression()
model.fit(X, y)
future_months = pd.DataFrame({
    "Month": range(len(monthly_sales), len(monthly_sales) + 3)
})
predictions = model.predict(future_months)
print("Future Sales Prediction:")
for i, value in enumerate(predictions, 1):
    print("Month", i, ":", round(max(0, value), 2))
plt.plot(monthly_sales.values, label="Actual Sales")
plt.plot(
    range(len(monthly_sales), len(monthly_sales) + 3),
    predictions,
    marker="o",
    label="Predicted Sales"
)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Future Sales Prediction")
plt.legend()
```

plt.show()

Output:



Result:
Future sales were successfully predicted using Linear Regression based on historical retail sales data. A graph was also generated to compare actual and predicted sales.